RAHUL

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Education

IIT (ISM) Dhanbad — Expected 2027
MTech. Fuel and Energy Engineering (CGPA 8.71/10) — *Dhanbad, Jharkhand*

- **Relevant Coursework:** Coal and Mineral Beneficiation, Biofuels, Fuel Technology, Computational Fluid Dynamics, Powerplant Engineering, Alternate Energy System, Energy Technology, Processing of Liquid and Gaseous Fuels, Mineral Processing Economics and Project Management, Research Methodology.

Government College of Engineering Salem — September 2023
B.E. Mechanical Engineering (CGPA 8.27/10) — *Salem, Tamil Nadu*

- **Relevant Coursework:** Thermodynamics, Fluid Mechanics, Heat Transfer, Engineering Mechanics, Strength of Material, Theory of Machines, Manufacturing Engineering, Industrial Engineering

Kendriya Vidyalaya No. 2 Port Blair — May 2018
Class XII (Percentage - 64.4) — *Port Blair, Andaman and Nicobar Islands*

- **Relevant Coursework:** Physics, Chemistry, Maths, English, Hindi

Kendriya Vidyalaya No. 2 Port Blair — May 2016
Class X (CGPA 9/10) — *Port Blair, Andaman and Nicobar Islands*

- **Relevant Coursework:** Mathematics, Science, SST, English, Hindi

Internships

Hindustan Shipyard Limited — Dec 2020 – Jan 2021
Intern — *Visakhapatnam, Andhra Pradesh*

- Designed Portable Jig in CATIA for the use in a ship under construction.
- Learned about General Assembly Drawings of ships.

Tata Steel Limited — May 2026 - Present
Research Intern — *Jamshedpur, Jharkhand*

- Proposed a novel dry beneficiation method for coal.
- Performed multiphase modelling and simulations using ANSYS Fluent 2024 R1.

Projects

Performance Analysis of Comfort Mattress using PCMs | *AutoCAD, CATIA v5, Ansys Fluent, LabVIEW, MS Excel, MS Word*

- Designed and simulated a cooling mattress under the environmental conditions of Tamil Nadu.
- With the use of thermocouples, collected temperature data at 12 heat zones on mattress using NI DAQ System and LabVIEW Software.
- Compared results of ordinary mattress and PCM integrated mattress.
- Developed a mattress to provide cooling by 25 percent.

Hydrodynamics studies of Vibrating Gas Fluidized Bed Separator for coal. | *Ansys Fluent, MS Excel, MS Word, Powerpoint*

- Did thorough literature review for recent advancements in CFD of fluidized bed hydrodynamics.
- Designed and simulated a VGFB Separator for hydrodynamics studies based on V.V. Buwa's work.
- Performed 2D transient multiphase eulerian simulation for model validation with previous work.
- Ran base case simulations without vibrational force term.
- Performed further simulations with vibrational force terms taken into the account.
- Optimized VGFB separator bed parameters and operating conditions.

Technical Skills

Languages: Python
Softwares: JupyterLab, AutoCAD, CATIA v5, Ansys Fluent, MS Excel, MS Word, Powerpoint, LabVIEW
Concepts: Computational Fluid Dynamics, Dry Beneficiation, Conventional Beneficiation, Powerplant Engineering, Coal Combustion, Life Cycle Assessment, Coal and Biomass Characterization, Crude Oil Characterization, Crude Oil Refining, Bio-fuel Production, Mineral Economics, Python Programming

Achievements

- GATE XE 2025 - AIR 1412
- AFCAT - Cleared 2 times
- CGCAT - Cleared 2 times

Social Engagements

Class Representative: MTech. Fuel and Energy Engineering, IIT (ISM) Dhanbad
Student Internship Representative : MTech. Fuel and Energy Engineering, IIT (ISM) Dhanbad
Anchoring: Teacher's Day 2025
Sports-Engagements: Cricket, Badminton, Volleyball
Conferences: INCOTHERM, N0ET, DRYBEN 2026